University of Utah — Spring 2026

# MATH 3210-004
# Midterm 1 Questions

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**Surname:** KEY

**First Name:**

**uNID:**

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1. **[10 points] Prove or disprove the following statement:** Let $f : \mathbb{R} \to \mathbb{R}$ be a function, $A, B \subseteq \mathbb{R}$ be two subsets. Then $f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B)$.

$(\subseteq)$ Let $x \in f^{-1}(A \setminus B)$. $\Rightarrow f(x) \in A \setminus B$

$\Rightarrow f(x) \in A$ but $f(x) \notin B$

$\Rightarrow x \in f^{-1}(A)$ but $x \notin f^{-1}(B)$

$\Rightarrow x \in f^{-1}(A) \setminus f^{-1}(B) \Rightarrow f^{-1}(A \setminus B) \subseteq f^{-1}(A) \setminus f^{-1}(B)$.

$(\supseteq)$ Let $x \in f^{-1}(A) \setminus f^{-1}(B)$. $\Rightarrow x \in f^{-1}(A)$, $x \notin f^{-1}(B)$

$\Rightarrow f(x) \in A$, $f(x) \notin B \Rightarrow f(x) \in A \setminus B$

$\Rightarrow x \in f^{-1}(A \setminus B) \Rightarrow f^{-1}(A) \setminus f^{-1}(B) \subseteq f^{-1}(A \setminus B)$.

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$\rightarrow S = \{q \in \mathbb{Q}_{\geq 0} \mid q^2 \leq 2\}$.

2. **[50 points]** Consider the set S that consists of all non-negative rational numbers x with the property that $x^2 \leq 2$.

(a) **[10 points] Prove or disprove the following statement:**
The supremum of S is $\sqrt{2}$.

$\cdot\ q \in S \Rightarrow q^2 \leq 2 \Rightarrow q \leq \sqrt{2} \Rightarrow \sqrt{2}$ is an upper bound.

$\cdot\ \varepsilon > 0$. Then by density of $\mathbb{Q}$,
$\exists q_\varepsilon \in \mathbb{Q} \cap ]\sqrt{2}-\varepsilon, \sqrt{2}[ \subseteq S$
$\Rightarrow \sqrt{2}$ is the least upper bound. □

(b) **[10 points] Prove or disprove the following statement:**
The maximum of S is $\sqrt{2}$.

We proved in class that $\sqrt{2} \notin \mathbb{Q}$. But $S \subseteq \mathbb{Q}$, so $\sqrt{2} \notin S$
$\Rightarrow \max(S)$ DNE. □

(c) **[10 points] Prove or disprove the following statement:**
The set of all lower bounds of S is $]-\infty, 0]$.

$\cdot$ Let $\lambda \in \mathbb{R}$ be a lower bound of $S$.
If $\lambda > 0$, by the density of $\mathbb{Q}$,
$\exists q_\lambda \in \mathbb{Q} \cap ]0, \tilde{\lambda}[ \subseteq S$, where
$\tilde{\lambda} = \min\{\lambda, \sqrt{2}\} \Rightarrow \lambda$ is not a lower bound!
$\Rightarrow \lambda \leq 0 \Rightarrow \lambda \in ]-\infty, 0]$.

Conversely, if $\lambda \in ]-\infty, 0]$, then $\lambda \leq 0 \leq s\ \forall s \in S \subseteq \mathbb{Q}_{\geq 0}$,
$\Rightarrow \lambda$ is a lower bound □.

(d) **[10 points] Prove or disprove the following statement:**
The smallest closed and bounded interval containing S is $[0, \sqrt{2}]$.

• $S = \{q \in \mathbb{Q}_{\geq 0} \mid q^2 \leq 2\} \subseteq [0, \sqrt{2}]$.

• Let $\lambda, \rho \in \mathbb{R}$, $\lambda < \rho$, be such that $S \subseteq [\lambda, \rho]$. Claim: $[0, \sqrt{2}] \subseteq [\lambda, \rho]$.

$\rho$ is an upper bound and $\lambda$ is a lower bound $\Rightarrow \sqrt{2} \leq \rho$ by (a) and $\lambda \leq 0$ by (c) $\Rightarrow [0, \sqrt{2}] \subseteq [\lambda, \rho]$. ✓

(e) **[10 points] Prove or disprove the following statement:**
The largest open and bounded interval contained in S is $]0, \sqrt{2}[$.

S contains no interval by Lemma.

Lemma (Density of Irrationals):
$\forall \alpha, \beta \in \mathbb{R}, \alpha < \beta \Rightarrow \exists \gamma \in \mathbb{R} \setminus \mathbb{Q}:$
$\alpha < \gamma < \beta$.

Pf:

By density of $\mathbb{Q}$, $\exists q \in \mathbb{Q} \cap ]\frac{\alpha}{\sqrt{2}}, \frac{\beta}{\sqrt{2}}[$
$\Rightarrow \gamma = q\sqrt{2} \in (\mathbb{R} \setminus \mathbb{Q}) \cap ]\alpha, \beta[$. D.

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3. **[20 points]** Let $x_\bullet$ be a sequence of real numbers and consider the following statements:

**A**: $x_\bullet$ converges to 0.

**B**: For any bounded sequence $b_\bullet$ of real numbers,

$$\lim_{n\to\infty} x_n b_n = 0.$$

(a) **[10 points] Prove or disprove the following statement:** If **A**, then **B**.

$$0 \leq |x_n b_n| = |x_n|\,|b_n| \leq \underbrace{|x_n|}_{\to 0}\underbrace{\left(\sup_n |b_n|\right)}_{<\infty} \xrightarrow{n\to\infty} 0$$

B.

(b) **[10 points] Prove or disprove the following statement:**
If **B**, then **A**.

$b_. : n \mapsto 1$ constant seq. $\Rightarrow$ bdd.

$\Rightarrow x_. = x_. b_. \to 0$ ✓

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4. **[15 points]** Let $x_\bullet : \mathbb{Z}_{\geq 1} \to \mathbb{R}$ be a sequence of real numbers, $x_* \in \mathbb{R}$ and consider the following statements:

**P**: $x_\bullet$ converges to $x_*$.

**Q**: The sequence of time averages of $x_\bullet$ converges to $x_*$, that is, $\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} x_k = x_*$.

(a) **[8 points]** **Prove or disprove the following statement:** If **P**, then **Q**.

$$\left|\frac{1}{n}\sum_{k=0}^{n-1} x_k - x_*\right| = \frac{1}{n}\cdot\left|\sum_{k=0}^{n-1}(x_k - x_*)\right|$$

$$\underset{\text{TI}}{\leq} \frac{1}{n}\sum_{k=0}^{n-1}|x_k - x_*|. \quad (*)$$

$P \Leftrightarrow \forall \varepsilon\ \exists N\ \forall n \geq N: |x_n - x_*| < \varepsilon. \ (\dagger)$

Let $\varepsilon > 0$ and $N$ be as $(\dagger)$. Then

$$\forall n \geq N: (*) = \frac{1}{n}\left(\sum_{k=0}^{N-1}|x_k - x_*| + \sum_{k=N}^{n-1}|x_k - x_*|\right)$$

$$\underset{(\dagger)}{\leq} \frac{1}{n}\left(\underbrace{\sum_{k=0}^{N-1}|x_k - x_*|}_{=M<\infty} + (n-1-N+1)\,\varepsilon\right) \xrightarrow[n\to\infty]{} \varepsilon$$

$\square$.

(b) **[7 points] Prove or disprove the following statement:**
If **Q**, then **P**.

Define $\forall n \geq 1: x_n = (-1)^n$.

$$\Rightarrow \sum_{k=0}^{n-1} x_k = \left\{ \begin{array}{l} -1, \text{ if } n \text{ is odd} \\ 0, \text{ if } n \text{ is even.} \end{array} \right].$$

| $n=$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| | -1 | 0 | -1 | 0 | -1 | 0 |

$$\Rightarrow \left| \frac{1}{n} \sum_{k=0}^{n-1} x_k \right| \leq \frac{1}{n} \longrightarrow 0.$$

$$\Rightarrow \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} x_k = 0.$$

But $x_\cdot$ is divergent (done in class).

$\square$.

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5. **[3 points]** Let $x_\bullet$ be a sequence of real numbers and $x_*$ be a real number. Consider the following statements:

**E**: $x_\bullet$ converges to $x_*$.

**F**: Any subsequence of $x_\bullet$ converges to $x_*$.

**G**: Any subsequence of $x_\bullet$ has a subsequence that converges to $x_*$.

(a) **[1 point] Prove or disprove the following statement:** If **E**, then **F**.

Let $\varepsilon > 0$. $\exists N \ \forall n \geq N : |x_n - x_*| < \varepsilon$.

Let $j$ be str. inc. Then

$\forall n \geq N : j(n) \geq n \geq N \Rightarrow |x_{j(n)} - x_*| < \varepsilon$

$\Rightarrow x_\bullet \circ j \to x_*$ ✓.

(b) **[1 point] Prove or disprove the following statement:** If **F**, then **G**.

Say $x_\bullet \circ j \to x_*$. By (a), any subseq of $x_\bullet \circ j$ also converges to $x_*$, and any seq. is a subseq of itself. $\square$

(c) **[1 point] Prove or disprove the following statement:** If **G**, then **E**.

Otherwise, $\exists$ $\boxed{x_\bullet \not\to x_*}$ (*) but any subseq. of $x_\bullet$ has a subseq. that converges to $x_*$.

(*) $\exists \varepsilon_0 > 0, \forall N, \exists n \geq N : |x_n - x_*| \geq \varepsilon_0$

Form: $N = 1 \to \exists n_1 \geq 1 : |x_{n_1} - x_*| \geq \varepsilon_0$

$N = n_1 + 1 \to \exists n_2 > n_1 : |x_{n_2} - x_*| \geq \varepsilon_0$

$N = n_2 + 1 \to \exists n_3 > n_2 : |x_{n_3} - x_*| \geq \varepsilon_0$

$\to$ By induction, $\exists$ str. inc. $j$:

$$\boxed{\forall n \geq 1 : |x_{j(n)} - x_*| \geq \varepsilon_0 > 0.}$$

This is a subseq., and since any further subseq of $x_\bullet \circ j$ also stays away from $x_*$, $x_\bullet \circ j$ has no subseq that converges to $x_*$, a contradiction!

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6. **[2 points]** Let $x_\bullet : \mathbb{Z}_{\geq 1} \to \mathbb{R}$ be a sequence of real numbers. Suppose $x_\bullet$ has the following property: for any $n, m \in \mathbb{Z}_{\geq 1}$,

$$x_{n+m} \leq x_n + x_m.$$

Then $\lim_{n\to\infty} \frac{x_n}{n}$ exists (possibly as an extended real number) and is equal to $\inf_{n\in\mathbb{Z}_{\geq 1}} \frac{x_n}{n}$.

$\boxed{\text{Lemma: } \forall n,m \in \mathbb{Z}_{\geq 1},\ \exists! q,r \in \mathbb{Z}_{\geq 0} : n = qm + r \text{ and } r < m.}$

Let $n,m \in \mathbb{Z}_{\geq 1}$. Then $x_n = x_{qm+r} \leq q \cdot x_m + x_r$ ⭐

$= (n-r)\cdot \frac{x_m}{m} + x_r \Rightarrow \boxed{\frac{x_n}{n} \leq \frac{n-r}{n}\cdot\frac{x_m}{m} + \frac{R}{n}}$

$\left(R = \max_{0\leq i<m} x_i < \infty\right)$

Taking limsup's over $n$'s, $\forall m \geq 1$:

$$\limsup_{n\to\infty} \frac{x_n}{n} \leq \frac{x_m}{m}.$$

$$\Rightarrow \limsup_{n\to\infty}\frac{x_n}{n} \leq \inf_{m\geq 1}\frac{x_m}{m} \leq \liminf_{n\to\infty}\frac{x_n}{n}$$

$$\Rightarrow \lim_{n\to\infty}\frac{x_n}{n} = \liminf_{n\to\infty}\frac{x_n}{n} = \limsup_{n\to\infty}\frac{x_n}{n} = \inf_n \frac{x_n}{n}$$ ✓

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Pf of Lemma: Induction on $n$.

$\boxed{n=1}$

$\boxed{m=1}$ XOR $\boxed{m>1}$

$\Downarrow$

$n = 1 = 1\cdot 1 + 0$

$\Rightarrow \boxed{q=1, \ r=0}$ ✓.

$n = 1 = 0\cdot m + 1$

$\Rightarrow \boxed{q=0, \ r=1}$ ✓.

Induction Step: Say ok for $n_* \in \mathbb{Z}_{\geq 1}$.

Thus $\forall m, \exists q_*, r_* : n_* = q_* m + r_*, \ r_* < m$.

$\Rightarrow$

$\boxed{r_* = m-1}$ XOR $\boxed{r_* < m-1}$

$\Downarrow$

$n_*+1 = q_* m + r_* + 1$

$= (q_*+1)m + 0$

$\Rightarrow \boxed{q = q_*+1, \ r = 0.}$

$\Downarrow$

$n_*+1 = q_* m + \underbrace{r_*+1}_{<m}$

$\Rightarrow \boxed{q = q_*, \ r = r_*+1}$

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Uniqueness :

Say $\bar{q}m+\bar{r} = n = \hat{q}m+\hat{r}$, $0\le\bar{r},\hat{r}<m$.

$0 \le |\hat{r}-\bar{r}| \le m-1$

Want : $\bar{q}=\hat{q}$, $\bar{r}=\hat{r}$.

$$0=(\bar{q}-\hat{q})m+(\bar{r}-\hat{r})$$

$$\Rightarrow |\hat{r}-\bar{r}| = |(\bar{q}-\hat{q})m| < m$$

$$\Rightarrow \boxed{\bar{q}=\hat{q}} \Rightarrow \boxed{\hat{r}=\bar{r}}$$

Pf of (b) : Induction on $q$ :

$$X_{(q+1)m+r} \le X_{(q+1)m} + X_r \le X_{qm}+X_m+X_r$$

$$\le qX_m+X_m+X_r = (q+1)X_m+X_r.$$

$\blacksquare$.